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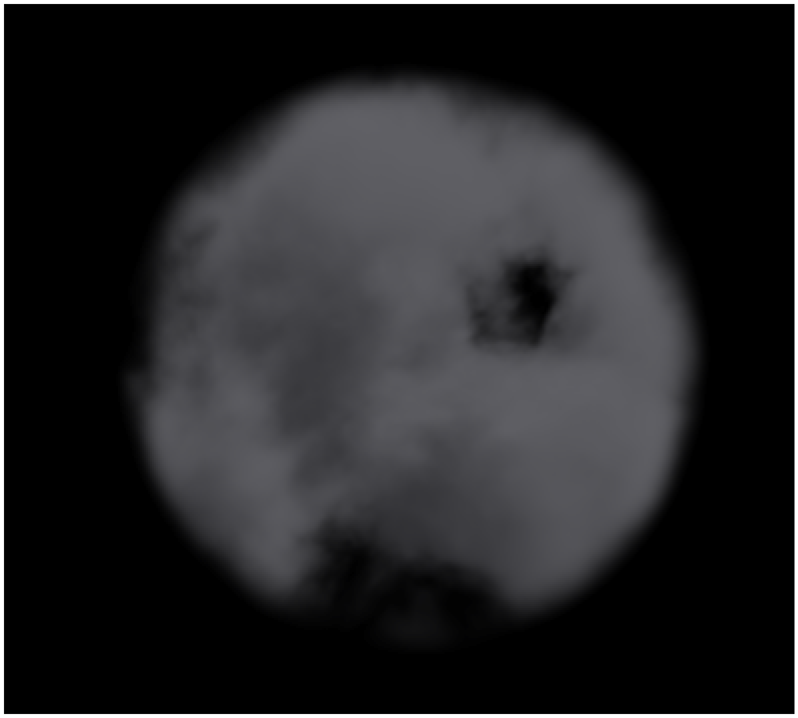
Project #8 FP



The goal of this project was to recreate the air scooter from Avatar: The Last Airbender, which is essentially a tornado contained inside a sphere shape. The vert file is simple, it passes vertex positions and camera ray data to the fragment shader for use in the time based movements. Raymarching was used to give the cloud ball volumetric depth, where a ray steps through the sphere and samples noise at each point to build up the illusion of a thick storm cloud. The noise is built from fractal Brownian motion with domain warping, which mathematically generates the wind pattern without using any textures. A pulse was added to shift the cloud density over time so it looks like a real changing wind, and a slow spin was added to the whole volume to keep it feeling alive. Instead of using the provided noise3 function the hash and noise functions were written manually, mostly because I forgot it existed. The main techniques are looping fractal Brownian motion and domain warping noise to produce a spherical tornado effect.

[Video Link]